Andres Buelvas

Full Stack Developer | API Integrations, n8n Automation and AI

Summary

Computer Engineer with 7+ years designing and delivering **scalable backend systems**, **fintech platforms**, and **AI-powered pipelines**. Specialized in **API integrations**, **workflow automation**, event-driven systems, and **LLM integration** using tools like LangChain, OpenAI, and Claude. Experienced in **leading engineering teams** and owning full product delivery cycles across fintech, document intelligence, and SaaS domains. Available for freelance projects in integrations, automation with n8n, AI features, and full stack development.

Experience

NextFaze **Apr 2025 – Jul 2026**

Backend Engineer | NestJS, FastAPI, LangChain, AWS, Claude

- Designed, implemented, and delivered the **AR System**, a middleware integration between Service Channel and FSG that automates the **full work order lifecycle** including requests, responses, invoicing, and notifications.
- Led the refactoring, implementation, testing, and maintenance of the **Document Parser**, enabling intelligent extraction, detection, and classification of **18+ document types** for ERP client integrations.
- Engineered integrations with **AI/OCR providers** (Datalab, DeepSeek, Mistral) to automate document information extraction from images and PDFs.
- Served as **primary decision-maker** for AI model and provider selection, evaluating Claude, GPT, Mistral, and DeepSeek against business model constraints, cost, and accuracy requirements.

SmartHop **Apr 2019 – Jan 2025**

Software Engineer IC5 / Engineer Manager | Node.js, React, MongoDB, GraphQL, Stripe ***Feb 2021 – Jan 2025***

- **Led a fintech engineering team** that scaled from 1 to 4 developers, responsible for designing, implementing, and delivering modules for invoicing, payroll, wallets, fuel, and billing.
- Co-designed and contributed to the implementation of **SmartPay**, an ACH-based payment mechanism serving **30–50 active SaaS clients**.
- Implemented **2FA enforcement** as a security requirement for UNIT API compliance, including bank account creation, debit card generation, and ACH/internal fund transfers.
- Architected a **double-entry ledger system** accurately recording income and expenses across client portfolios handling up to **70 records per carrier**.
- Led **Payroll V2** redesign with the full team, delivering the module within a **single quarter** with on-demand income/expense generation and PDF reporting.
- Integrated **Onramp** for fuel card generation and transaction visualization.
- Implemented gas station **geolocation mapping** using GraphQL, Google Maps, and Trimble Maps API.

Senior Full Stack Engineer | Node.js, React, Redis, Plaid, Stripe ***Aug 2020 – Feb 2021***

- **Sole developer** responsible for all e-commerce, fuel, and insurance products across the platform.
- Integrated **EFS/WEX API** via SOAP for fuel card generation and transaction tracking, serving 5–10 users under an OOP architecture.
- Integrated **Plaid OAuth** to enable secure bank authentication across **20–30 connected banking institutions**.
- Designed and delivered **Payroll V1** infrastructure **end-to-end in under 3 months**.

Full Stack Engineer | Node.js, Python, React, MongoDB, Stripe ***Apr 2019 – Aug 2020***

- Built an interactive **profit dashboard** using Chart.js and React for an early client base of **5–15 active accounts**.
- Integrated **Stripe** for subscription plan management and secure payment handling across **10–100 active users** including free trial conversions.
- Implemented geolocation queries with **Redis set operations** for efficient resource retrieval.
- Developed transactional email flows using **Mandrill** and **Mailchimp**.

CIDITEL–VE / Latinux Inc **Jul 2018 – Apr 2019**

Backend Engineer | Python, Web2py

- Built the **MECE system**, an alternative admission pathway for students applying to Simon Bolivar University.
- Implemented a **timed online academic exam platform** with separate categories for numerical and verbal ability.
- Enabled **real-time exam control** for teachers, including session initiation and instant results feedback.

Simon Bolivar University **Apr 2018 – Jul 2018**

Support Assistant | Python, Web2py

- Built **SisPIO**, a platform that helped students prepare for university entrance exams.
- Supported administrative operations including **student tracking** and PIO program activity management.

Projects

EzzyGas

Fuel Management & Fintech | NestJS, Next.js, React Native, PostgreSQL, Redis, AWS, Docker, Stripe, Plaid

- Designed a **multi-app monorepo** (backend, mobile, admin web, station gateway) with shared TypeScript types enforcing API contracts across all services.
- Built a modular backend with **30+ domain-driven modules**, repository pattern, **ABAC authorization**, and workflow orchestration via **XState state machines**.
- Developed a **multi-tenant financial system** with an immutable append-only ledger, account-group-scoped wallets.
- Built a **cross-platform mobile app** with biometrics, geolocation, and push notifications.
- Integrated **Stripe Issuing** for virtual/physical fuel cards, **Plaid** for bank linking and ACH funding, and **Marqeta** for digital wallet provisioning.
- Implemented **real-time** backend to fuel pump hardware communication via Redis command queues and WebSockets.
- Provisioned cloud infrastructure with **CloudFormation** (ECS Fargate, ECR, ElastiCache) and **CI/CD** via GitHub Actions + OIDC with zero static credentials.

Document Parser

AI Document Processing Pipeline | Python, FastAPI, PostgreSQL, OpenAI, Claude, Langfuse, AWS, Docker, DataLab

- Built an **AI-powered document processing pipeline** supporting **18 document types** using a **10-stage orchestration architecture** with the Strategy Pattern.
- Designed a **hybrid LLM classification system** combining universal and binary classifiers, achieving **over 90% accuracy** on complex multi-document PDFs.
- Implemented a **bidirectional graph-based algorithm** to detect and link supporting documents across PDF pages, with conflict resolution to eliminate redundant LLM calls, including full retry/reclassification flow preserving document associations across pipeline re-runs.
- Built a **self-training loop** with Langfuse datasets, **LLM-as-Judge** evaluators, and automated prompt experimentation using candidate vs. baseline comparison.
- Implemented **AWS Lambda** email ingestion pipeline (SES to S3 to Parser API) with attachment extraction and multi-environment routing.
- Structured the codebase to support an **AI-assisted development framework** with modular rules, reusable skills, and specialized sub-agents integrated into **GitHub Actions workflows**, enabling autonomous implementation of new document types.

Account Receivable (AR) System

Enterprise Systems Integration | NestJS, TypeScript, RabbitMQ, Prisma, PostgreSQL, Redis, Docker, Datadog

- Built a **bidirectional AR integration system** connecting a facility management provider (ServiceChannel) with an ERP (FSG), automating work order data flow end-to-end.
- Designed the architecture using **Hexagonal Architecture** and **event-driven patterns**, keeping business logic decoupled from external services and infrastructure.
- Implemented complex **multi-step workflows** with XState state machines across inbound, core, and outbound processing layers.
- Engineered reliable **async messaging** with RabbitMQ including retry policies, **dead letter queues**, and **idempotent message processing**.
- Built a **real-time monitoring dashboard** with queue management, DLQ replay, and message lifecycle tracking.
- Added observability via **Datadog APM** with custom metrics, distributed tracing, and structured logging.

Photo Post

Social Media Microservices API | Node.js, Express.js, SQLite, Redis, Kafka, Docker, JWT, Jest

- Built a REST API with **5 independent microservices** (auth, users, posts, likes, gateway) with an **API Gateway** centralizing routing, JWT authentication, and internal request signing via x-gateway-signature headers.
- Implemented **Kafka** for asynchronous event-driven communication and **per-service Redis caching** for token blacklisting and response caching with 5-min TTL.
- Stored and processed user-uploaded images using Multer and Sharp, containerized with **Docker Compose** including dedicated Redis instances and SQLite volumes per service.
- Wrote **unit and integration tests** with Jest across all services.

CQRS Blog Application

Microservices Blog API | Node.js, TypeScript, React, Express.js, MySQL, MongoDB, RabbitMQ

- Built a **CQRS-based blog platform** using a microservices architecture with an **API Gateway** as single entry point.
- Routed read operations to a Queries Service backed by **MongoDB** and write operations to a Commands Service backed by **MySQL**, enforcing strict **read/write model separation**.
- Implemented **event-driven synchronization** between services via RabbitMQ, where write operations publish domain events that update the read model asynchronously.
- Built a **React + TypeScript** frontend with Vite and Shadcn/UI consuming the gateway API.

ClueGarden

SaaS Subscription Platform | Node.js, Express.js, Firebase, Stripe, Heroku

- Built a **backend payments and subscription management service** integrating Stripe and Firebase for a SaaS platform.
- Integrated **Stripe Checkout**, Billing Portal, and **webhook event processing** for full subscription lifecycle management including creation, updates, and cancellations, secured via **webhook signature verification**.
- Mirrored Stripe objects into **Firestore** for low-latency reads and audit trail, with **automated free trial expiration** via daily cron job.
- Implemented **Firebase Authentication** with JWT middleware for all protected endpoints across a component-based REST API separating users, billing, and payments modules.

Portfolio

Personal Portfolio | Next.js, React, TypeScript, Tailwind CSS, Anthropic SDK, Zod, Vercel

- Built a production-grade personal portfolio with **Next.js App Router** using a **hybrid SSR/CSR architecture** to maximize performance while keeping UI interactions client-side.
- Integrated an **AI-powered chat assistant** using **Claude Haiku** with tool-calling capabilities, fetching CV content, GitHub profile data, and performing scoped web searches in real time.
- Implemented a **custom i18n system** (EN/ES) without external libraries using React Context and cookie-based persistence for language and theme preferences.
- Built a **secure contact form** with Zod validation, HTML sanitization, and tiered in-memory rate limiting per-IP and global, with **prompt injection protection** on the AI layer.
- Designed a **custom design system** with Tailwind CSS custom tokens, dark/light mode via CSS variables, and custom React hooks for typewriter effects, infinite carousels, and responsive breakpoints.
- Used **Google Stitch** to prototype UI mockups and page layouts prior to implementation.

Education

Simon Bolivar University — **2013 – 2019**
Bachelor in Technology (BTech) in Computer Engineering — ***Caracas, Venezuela***

Pontificia Universidad Javeriana — **2025**
Diploma in Software Architecture and Cloud Computing — ***Cali, Colombia***

Coursework

Modern Software Architecture: DDD, Events, Microservices — *Apr 2025*

Master in Microservices with JavaScript and Node.js *Mar 2025*
AWS Cloud Technical Essentials *Jan 2025*
EF SET Certificate *Jan 2025*
Programming in Python *Apr 2024*
Clean Code *Mar 2024*

Languages

Spanish — Native
English — B2 (Professional Working Proficiency)

Skills

Languages: JavaScript, TypeScript, Python

Backend: Node.js, Express.js, NestJS, FastAPI, Kafka, RabbitMQ, SQS, Web2py

Frontend: React, Next.js, Remix, Redux, Vite, Tailwind CSS, Shadcn/UI, HTML5, CSS, Material UI, Bootstrap

AI & LLM: LangChain, OpenAI, Claude, Mistral AI, Langfuse, MCP, Prompt Engineering

Automation & CMS: n8n, WordPress, Shopify, Wix

Testing: Jest, Playwright, Puppeteer, Pytest

Databases: MongoDB, MySQL, PostgreSQL, Redis, Firebase, Supabase, DynamoDB, SQLite

Cloud & DevOps: AWS (EC2, ECS, ECR, Fargate, Lambda, S3, SQS, API Gateway, CloudWatch, CloudFormation), Google Cloud, Docker, GitHub Actions, Bitbucket Pipelines

Developer Tools: VSCode, Cursor, Windsurf, Postman, Insomnia, Swagger, Datadog, Git

Project Management: Jira, Linear, Bitbucket, GitHub

Concepts: OOP, Microservices Architecture, Event-Driven Architecture, Domain-Driven Design (DDD), CQRS, Hexagonal Architecture, MVC, Repository Pattern, Strategy Pattern, Design Patterns, REST API, GraphQL, CI/CD, Agile, Auto Scaling, Load Balancing, Serverless, Real-Time Applications, State Machines, Multi-tenancy, Idempotency, Dead Letter Queues, ABAC Authorization, OAuth, Webhook Processing, LLM-as-Judge